Deva Anand

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Education

University of Massachusetts Amherst — **September 2025 – May 2027 (Expected)**
Master of Science in Computer Science — Amherst, MA
Coursework: Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies — **August 2019 – May 2023**
Bachelor of Engineering in Computer Science and Engineering — Chennai, India

Skills

Languages	Python, JavaScript, TypeScript, Java, C++, SQL, Bash, HTML/CSS
Backend & APIs	Node.js, Express, FastAPI, Fastify, REST APIs, async services, input validation, authentication, rate limiting
Testing & Quality	Integration testing, test automation (pytest), regression testing, test-case development, code review
Cloud & DevOps	AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux
Databases & Web	MongoDB, PostgreSQL, Redis, SQLite, React, Electron, Webpack

Experience

Wysa — **July 2023 – July 2025**
Backend Engineer — Bengaluru, India

- Engineered a **multi-tenant NHS eTriage backend** (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly patient submissions** as a production system integrated with Mayden's iaptus platform via REST APIs and aggregation pipelines; contributed to **$340K+** in revenue.
- Participated in the full software-quality process across the eTriage stack: developed and executed **integration test suites** for regression protection, reviewed code, monitored production logs during releases, and remediated **20+ VAPT findings** to safeguard sensitive clinical data.
- Partnered with the AI team to gather requirements and architect a full-stack data annotation platform (**React, Node.js, MongoDB**), replacing manual spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved throughput by **60%**.

Cario Growth Services Pvt. Ltd. — **November 2022 – May 2023**
Backend Developer Intern — Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js and Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories, tightening type safety and API contracts to improve maintainability and reduce defects.

Projects

AgenticSearch - Production-Grade Multi-Stage Service — github.com/Deva-1903/ciir_agentic_search
Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage Python/FastAPI service that turns open-ended queries into structured, verified results through typed planning, deterministic parsers with fallback, output verification, and cell-level provenance for every value.
- Hardened the service with **150+ automated tests**, provider routing and fallback, pipeline instrumentation, and a reviewer-facing UI, intentionally engineered for reliability, reproducibility, and maintainability.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App — github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare
React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **cross-platform desktop application** integrating camera, microphone, frontend UI, and backend services into a single on-device pipeline, prototyped, designed, and shipped end-to-end under hackathon time pressure.
- Engineered a **FastAPI backend** and **Electron/React frontend** that run fully offline with no external network dependencies; designed async APIs and queues to keep the UI responsive while handling local video and audio streams.

Spark ETL Performance Optimization (NYC TLC Trip Data) — github.com/Deva-1903/Spark-ETL-Optimization
Apache Spark / PySpark, Databricks, SQL, Python

- Optimized a distributed Spark ETL pipeline by tuning broadcast joins, caching, partitioning, and adaptive settings; profiled stage-level timings via Spark UI to localize shuffle and skew bottlenecks across naive vs. tuned runs.
- Documented a repeatable, reviewable optimization workflow over the same dataset, capturing column-pruning and precomputed-field trade-offs so configuration choices stay analyzable for downstream jobs.

Publication & Extracurriculars

- **Publication:** *"Alzheimer's Disease Classification using Transfer Learning,"* **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.